Solution / Buns

2110201: COMP ENG MATH (2023/2) Quiz 4

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1. (5 points) Given a matrix $\mathbf{A} = \begin{bmatrix} 2 & -2 & 6 & 6 \\ 2 & -1 & 5 & 8 \\ 3 & -4 & 10 & 2 \end{bmatrix}$

1.1. (3 points) Find one matrix $\mathbf{B}$ such that $\text{col}(\mathbf{B}) = \text{nul}(\mathbf{A})$.

First, we will try to find nul(A) in a "span" term.
(As "col" is defined with "span")
nul(A) = perp of rows → convert to span.

$$\begin{bmatrix} 2 & -2 & 6 & 6 \\ 2 & -1 & 5 & 8 \\ 3 & -4 & 10 & 2 \end{bmatrix} \overset{\text{RRE}}{\Rightarrow} \begin{array}{c} \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ \begin{matrix} P & P & F & P \end{matrix} \end{array}$$

$\therefore$ nul(A) = span $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

Then, we can take $B = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

such that col(B) has that exact span.

1.2. (2 points) Find one set of vectors (called S) such that $\text{span}(S) = \text{col}(\mathbf{A})$ and each vector in S is linearly independent.

From 1.1, the free column of A is dependent, and is a linear combination of other vectors in the pivot columns. $\therefore S = \left\{ \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 8 \\ 2 \end{bmatrix} \right\}$

You can also answer $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$!

2. (3 points, 1 point each) Provide short answer.

2.1. Given a set of vectors $V = \left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. Is V independent? Give us a short explanation.

Yes, $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \\ 4 & 3 & 1 \end{bmatrix} \overset{\text{RRE}}{\Rightarrow} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ every column is a pivot column. #

2.2. Find a plane that contains $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -2 \end{bmatrix}$ and is also a subspace in $\mathbf{R}^3$ (Answer in any form, just make it understandable.)

Solution 1

Since this plane is a subspace, it must contain the origin.
You can use the strategies in Unit 3 to solve for the equation. (Quiz 2) #

Solution 2

It is obvious (and not at the same time) that the plane is span $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ -2 \end{bmatrix} \right\}$ #

2.3. Consider the statement: "If $\text{row}(\mathbf{A}) = \text{col}(\mathbf{A})$, then $\mathbf{A}$ is invertible." Is the statement true? Give us a short explanation.

False $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ contradicts this statement. #

3. (2 points) Let m, n be positive integers. Given an $n \times m$ matrix $\mathbf{A}_{n\times m}$ and an $n \times n$ matrix $\mathbf{B}_{n\times n}$.
Prove that $\text{nul}(\mathbf{A}) \subseteq \text{nul}(\mathbf{BA})$. [**Hint:** Prove that if $\mathbf{x}$ is a vector such that $\mathbf{x} \in \text{nul}(\mathbf{A})$, then $\mathbf{x} \in \text{nul}(\mathbf{BA})$]

Let $\mathbf{x} \in \mathbb{R}^m$ such that $\mathbf{x} \in \text{nul}(A)$

From the test of membership of nullspace, $A\mathbf{x} = \mathbf{0}$ ($\mathbf{0}$ is a zero vector in $\mathbb{R}^n$)

Then $(BA)\mathbf{x} = B(A\mathbf{x}) = B(\mathbf{0}) = \mathbf{0}$

$\therefore \mathbf{x} \in \text{nul}(BA)$

This proves the statement: $\text{nul}(A) \subseteq \text{nul}(BA)$

Note: can you do the reverse? ($\text{nul}(BA) \subseteq \text{nul}(A)$)
what other information is needed?